Truth Tables with Three Statements

When constructing a truth table involving three simple statements, we use the variables p, q, and r. Adding a third statement increases the number of possible combinations to eight.

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

When constructing a truth table:

- List all possible combinations of truth values.
- Evaluate any negations (∼) of individual statements first.
- Evaluate the expressions inside parentheses.
- Evaluate the remaining compound statement to determine its final truth value.

Note

If a negation applies to an entire expression in parentheses, evaluate the expression inside the parentheses before applying the negation.

Remember that AND (∧) is true only when both statements are true, while OR (∨) is true when at least one statement is true.

Examples

Construct a truth table for the following compound statements.

a) $p \vee (q \wedge r)$

p	q	r		

b) $p \wedge (q \vee \sim r)$

p	q	r			

c) $(\sim p \vee \sim q) \wedge r$

p	q	r				

d) $(p \wedge \sim q) \vee \sim r$

p	q	r				

5.1 Discuss with examples

5.1.3 Conditional statement

Many statements in everyday language take the form "If p, then q." For example,

- If I study, then I will pass the test.
- If it rains tomorrow, then I will stay home.

In logic, these are called conditional statements.

A conditional statement is a compound statement in the form "If p, then q."
It is written symbolically as

$$p \rightarrow q$$

where p is the hypothesis (the "if" part), and q is the conclusion (the "then" part).

Examples

Consider the following simple statements:

p: It is sunny.

q: I am going to the park.

Write each of the following using symbolic logic.

a) If it is sunny, then I am going to the park.

b) If it is not sunny, then I am not going to the park.

c) If I am going to the park, then it is sunny.

The statement in part (c) is called the converse of the original conditional statement.

The converse is formed by reversing the order of the two statements.

Original conditional: $p \rightarrow q$

Converse: $q \rightarrow p$

Note

The converse is not necessarily true simply because the original conditional statement is true.

5.1.4 Biconditional statement

A biconditional statement is a compound statement in the form “p if and only if q.”

It is written symbolically as

$$p \leftrightarrow q$$

A biconditional statement combines a conditional statement and its converse.

In other words, both of the following must be true:

- If p, then q.
- If q, then p.

Consider the statement:

"A whole number is even if and only if it is divisible by 2."

Let:

p: A whole number is even.

q: The whole number is divisible by 2.

The biconditional statement is: $p \leftrightarrow q$

This statement is true because both the conditional statement and its converse are true.

Examples

Consider the following simple statements:

p: A number is positive.

q: The number is greater than zero.

Write each statement using symbolic logic.

a) If a number is positive, then it is greater than zero.

b) A number is positive if and only if it is greater than zero.

c) If a number is greater than zero, then it is positive.

Determine whether each biconditional statement is true or false. Explain your answer.

a) An integer is even if and only if it is divisible by 2.

b) An integer is divisible by 4 if and only if it is even.

c) A real number is positive if and only if its square is positive.

5.1.5 Inverse statement

The inverse of a conditional statement is formed by negating both the hypothesis and the conclusion without changing their order.

If the original conditional statement is:

$$p \rightarrow q$$

then its inverse is:

$$\sim p \rightarrow \sim q$$

Consider the conditional statement:

"If a number is divisible by 4, then it is even."

The inverse is:

"If a number is not divisible by 4, then it is not even."

Note

The inverse is not necessarily true simply because the original conditional statement is true.

For example, 6 is not divisible by 4, but it is even.

Examples

Write the inverse of each conditional statement.

a) If it rains, then the ground is wet.

b) If a number is divisible by 10, then it is divisible by 5.

c) If a computer is connected to the Internet, then it can access online resources.